# Project Report - Day 3

## Title

TASK-03: Developing a Keyword-Based Search & Ranking Engine

## Objective

To build a simple keyword-based search engine that preprocesses text and ranks sentences based on their relevance to a user query using Jaccard similarity.

## Libraries Used

1. **String**
    - Used to remove punctuation from text during preprocessing to ensure accurate word matching.

## Errors

1. **Incorrect Corpus File Path**
    - **Issue:** If corpus.txt is not in the same folder as the script, the program may raise a file not found error.
    - **Solution:** Keep corpus.txt in the same directory as the script or provide the correct full path.
2. **Punctuation Affecting Search Accuracy**
    - **Issue:** Words with punctuation such as code. and code were treated differently.
    - **Solution:** Punctuation was removed using string.punctuation before token comparison.
3. **Common Stopwords Reducing Search Quality**
    - **Issue:** Words like the, is, in, and of created noise in similarity matching.
    - **Solution:** A stopword list was added and such words were removed during preprocessing.

4. **Empty Query or Empty Token Set Problem**

- **Issue:** If the query or processed sentence became empty, similarity calculation could cause invalid division.
- **Solution:** A condition was added in jaccard_similarity() to return 0 when the union is empty.

## Output

```
0.0 -> 5.       Large Language Models require massive amounts of clean data for fine-tuning.
PS D:\Daily_Tasks\Day3_Task> & C:\Users\Acer\anaconda3\envs\mlenv\python.exe d:/Daily_Tasks/Day3_Task/ta
y
Enter your search query: machine learning



0.22 -> 9.      Python is the primary language used for data science and machine learning.
0.09 -> 20.     Learning to play a musical instrument can improve cognitive function in children.
0.09 -> 28.     PyTorch and TensorFlow are the two most popular frameworks for deep learning.
0.08 -> 34.     Reinforcement Learning from Human Feedback (RLHF) helps align AI with values.
0.08 -> 22.     Zero-shot learning allows a model to recognize objects it has never seen before.
PS D:\Daily_Tasks\Day3_Task> & C:\Users\Acer\anaconda3\envs\mlenv\python.exe d:/Daily_Tasks/Day3_Task/ta
y
Enter your search query: ai models

Top Results:

0.09 -> 5.      Large Language Models require massive amounts of clean data for fine-tuning.
0.09 -> 42.     BERT and RoBERTa are encoder-only models used for understanding text context.
0.08 -> 30.     Hallucinations occur when an AI model generates confident but false information.
0.08 -> 34.     Reinforcement Learning from Human Feedback (RLHF) helps align AI with values.
0.08 -> 36.     GitHub Copilot uses AI to suggest code completions in real-time as you type.
PS D:\Daily_Tasks\Day3_Task>
```